

DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK II

III EXACT DIFFERENTIAL EQUATIONS

Definition Let F be a function of class C^1 in a domain D .

The total diff of F of the function F is defined by the formula

$$dF(x,y) = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy \quad \text{for all } (x,y) \text{ in } D.$$

Ex 7 Obtain the total dF of the function $F(x,y) = 5x^3y + y^3 \sin x^2$

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$$

$$dF = (15x^2y + \cos(x^2)y^3 \cdot 2x) dx + (5x^3 + 3y^2 \sin x^2) dy$$

Ex 7 Show that $f(x,y) = x^2y^2$.

$$3x^2y^2 dx + 2x^3y dy = dF \text{ is the total diff of}$$

$$\frac{\partial f}{\partial x} = 3x^2y^2 \quad \frac{\partial f}{\partial y} = 2x^3y$$

Theorem Consider the differential equation $M(x,y)dx + N(x,y)dy = 0$ where M and N of class C^1 (has cont. derivatives) in a domain D . The differential equation

$$M(x,y)dx + N(x,y)dy = 0$$

is exact. if and only if $\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}$ for all $(x,y) \in D$

HOW TO SOLVE EXACT DIFFERENTIAL EQUATIONS?

① Check if $\frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}$ "Exactness condition"

② If ① is satisfied $F(x,y) = \int M(x,y) dx + h(y)$ and find $h(y)$ as follows

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y} \int M(x,y) dx + h'(y) = N(x,y)$$

OR

$F(x,y) = \int N(x,y) dy + g(x)$ and find $g(x)$ as follows

$$\frac{\partial F}{\partial x} = \frac{\partial}{\partial x} \int N(x,y) dy + g'(x) = M(x,y)$$

$F(x,y)$ is called potential function

Ex ① $(6xy + 2y^2 - 5) + (3x^2 + 4yx - 6) \frac{dy}{dx} = 0$

$$\underbrace{(6xy + 2y^2 - 5)}_{M(x,y)} dx + \underbrace{(3x^2 + 4xy - 6)}_{N(x,y)} dy = 0$$

First check if it is separable, not sep. $2y^2 dx, 3x^2 dy$ not sep.

$$M_y = 6x + 4y$$

$$N_x = 6x + 4y$$

$$M_y = N_x \checkmark$$

$$\begin{aligned} \textcircled{a} \quad F(x,y) &= \int (6xy + 2y^2 - 5) dx + h(y) \\ &= 3x^2y + 2xy^2 - 5x + h(y) \end{aligned}$$

$$\frac{\partial F}{\partial y} = 3x^2 + 4xy + h'(y) \stackrel{?}{=} N(x,y)$$

$$\cancel{3x^2} + \cancel{4xy} + h'(y) = \cancel{3x^2} + \cancel{4xy} - 6$$

$$h'(y) = -6 \quad h(y) = -6y$$

$$\boxed{F(x,y) = 3x^2y + 2xy^2 - 5x - 6y = C}$$

$$\textcircled{b} F(x,y) = \int (3x^2 + 4xy - 6) dy + g(x)$$

$$= 3xy + 2y^2 - 6y + g(x)$$

$$\frac{\partial F}{\partial x} = 3y + 2y^2 + g'(x) = 6xy + 2y^2 - 6$$

$$g'(x) = -6 \quad g(x) = -6x$$

$$\boxed{F(x,y) = 3x^2y + 2xy^2 - 6y - 6x = C.}$$

$$\textcircled{2} (ye^x + 2e^x + y^2) dx + (e^x + 2xy) dy = 0 \quad y(0) = 6$$

NOT SEP. $y^2 dx, e^x dy$

NOT HOM. $e^x dy$

$$M(x,y) = ye^x + 2e^x + y^2$$

$$M_y = N_x \quad \begin{cases} \frac{\partial M}{\partial y} = e^x + 2y \\ N(x,y) = e^x + 2xy \\ \frac{\partial N}{\partial x} = e^x + 2y \end{cases}$$

$$F(x,y) = \int (e^x + 2xy) dy + g(x) = e^x y + xy^2 + g(x)$$

$$\frac{\partial F}{\partial x} = e^x y + y^2 + g'(x) = M(x,y) = e^x y + 2e^x + y^2$$

$$g'(x) = 2e^x \quad g(x) = 2e^x + C$$

$$e^x y + xy^2 + 2e^x = C \rightarrow \text{Gen. sol.}$$

$$\underline{y(0)=6} \quad e^0 y(0) + 0 \cdot y(0) + 2 \cdot e^0 = C$$

$$6 + 2 = C \quad C = 8$$

Particular solution $e^x y + xy^2 + 2e^x = 8$

$$\textcircled{3} (3x^2 + 3xy^2) dx + (3x^2y - 3y^2 + 2y) dy = 0$$

$$\underbrace{(3x^2 + 3xy^2)}_{M(x,y)}$$

$$\underbrace{(3x^2y - 3y^2 + 2y)}_{N(x,y)}$$

NOT SEP. $3xy^2 dx$
NOT HOM. $3x^2y, 3y^2$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$6xy = 6xy \quad \checkmark$$

$$F(x,y) = \int (3x^2 + 3xy^2) dx + h(y) = x^3 + \frac{3}{2} x^2 y^2 + h(y)$$

$$\frac{\partial F}{\partial y} = 3x^2 y + h'(y) = N(x,y) = 3x^2 y - 3y^2 + 2y$$

$$h'(y) = -3y^2 + 2y$$

$$h(y) = -y^3 + y^2 + C.$$

$$F(x,y) = x^3 + \frac{3}{2} x^2 y^2 - y^3 + y^2 = C.$$

EXERCISES

$$\textcircled{1} \underbrace{(\sin x + y) dx}_{M(x,y)} + \underbrace{(x - 2\cos y) dy}_{N(x,y)} = 0$$

NOT SEP. $y dx, x dy$
NOT HOM. $\sin x, \cos y$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 1 \quad \text{EXACT}$$

$$\frac{\partial F}{\partial x} = \sin x + y$$

$$\frac{\partial F}{\partial y} = x - 2\cos y$$

$$F(x,y) = \int (\sin x + y) dx + h(y) = -\cos x + xy + h(y)$$

$$\frac{\partial F}{\partial y} = x + h'(y) = x - 2\cos y$$

$$h'(y) = -2\cos y \quad h(y) = -2\sin y + C$$

$$\underbrace{-\cos x + xy - 2\sin y = C.}$$

$$\textcircled{2} \underbrace{(ye^{xy} + 3x^2) dx}_{M(x,y)} + \underbrace{(xe^{xy} - \cos y) dy}_{N(x,y)} = 0.$$

$$\frac{\partial M}{\partial y} = e^{xy} + xye^{xy}$$

$$= \frac{\partial N}{\partial x} = e^{xy} + xye^{xy} \quad \checkmark \quad \text{EXACT}$$

$$F(x,y) = \int (ye^{xy} + 3x^2) dx + h(y) = e^{xy} + x^3 + h(y)$$

$$\frac{\partial F}{\partial y} = e^{xy} x + h'(y) = xe^{xy} - \cos y \quad h'(y) = -\cos y$$

$$h(y) = -\sin y + C$$

$$F(x,y) = e^{xy} + x^3 - \sin y = C.$$

$$(3) \quad \underbrace{(6xy^2 - 3x^2)}_{M(x,y)} dx + \underbrace{(6x^2y + 3y^2 - 7)}_{N(x,y)} dy = 0$$

$$\frac{\partial M}{\partial y} = 12xy = \frac{\partial N}{\partial x} = 12xy \quad \checkmark$$

$$F(x,y) = \int (6xy^2 - 3x^2) dx + h(y) = 3x^2y^2 - x^3 + h(y)$$

$$\frac{\partial F}{\partial y} = 6x^2y + h'(y) = 6x^2y + 3y^2 - 7 \quad h'(y) = 3y^2 - 7 \quad h(y) = y^3 - 7y$$

$$F(x,y) = 3x^2y^2 - x^3 + y^3 - 7y = C$$

$$(4) \quad \underbrace{(2x^3 - xy^2 - 2y + 3)}_{M(x,y)} dx - \underbrace{(x^2y + 2x)}_{N(x,y)} dy = 0$$

$$\frac{\partial M}{\partial y} = -2xy - 2 = \frac{\partial N}{\partial x} = -2xy - 2$$

$$F(x,y) = \int (2x^3 - xy^2 - 2y + 3) dx + h(y) \\ = \frac{x^4}{4} - \frac{xy^2}{2} - 2xy + 3x + h(y)$$

$$\frac{\partial F}{\partial y} = -\cancel{x^2y} - \cancel{2x} + h'(y) = -\cancel{x^2y} - \cancel{2x} \\ h'(y) = 0 \quad h(y) = C$$

$$F(x,y) = -\frac{x^2y^2}{2} - 2xy + \frac{x^4}{2} + 3x = C$$

$$(5) \quad 2xy dx + x^2 dy = 0$$

$$\text{SEP.} \quad \frac{dy}{dx} = -\frac{2xy}{x^2} \quad \frac{dy}{-2y} = \frac{dx}{x}$$

$$\text{HOM.} \quad f(x,y) = -\frac{2xy}{x^2} = -\frac{2y}{x}$$

$$f(x, \alpha y) = -2 \frac{\alpha x \alpha y}{\alpha^2 x^2} = -\frac{2xy}{x^2} = f(x,y)$$

$$\underline{\text{EXACT}} \quad M(x,y) = 2xy \quad N(x,y) = x^2$$

$$\frac{\partial M}{\partial y} = 2x \quad \frac{\partial N}{\partial x} = 2x \quad \checkmark$$

(5)

$$F(x,y) = \int 2xy \, dx + h(y)$$

$$F(x,y) = x^2y + h(y) \quad \frac{\partial F}{\partial y} = x^2 + h'(y) = x^2 \quad h'(y) = 0 \quad h(y) = C$$

$$F(x,y) = x^2y = C$$

$$\textcircled{6} \underbrace{(e^x + 2xy^2)}_{M(x,y)} dx + \underbrace{(\cos y + 3x^2y^2)}_{N(x,y)} dy = 0$$

$$\frac{\partial M}{\partial y} = 2xy^2 = \frac{\partial N}{\partial x} = 2xy^2 \quad \checkmark$$

$$F(x,y) = \int (e^x + 2xy^2) dx + h(y)$$

$$= e^x + x^2y^2 + h(y)$$

$$\frac{\partial F}{\partial y} = 2x^2y + h'(y) = \cos y + 2x^2y \quad h'(y) = \cos y \quad h(y) = \sin y$$

$$F(x,y) = e^x + x^2y^2 + \sin y = C$$

$$\textcircled{7} \underbrace{(2xy + 1)}_{M(x,y)} dx + \underbrace{(6y + x^2)}_{N(x,y)} dy = 0$$

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x} = 2x \quad \checkmark$$

$$F(x,y) = \int (2xy + 1) dx + h(y) = x^2y + x + h(y)$$

$$\frac{\partial F}{\partial y} = x^2 + h'(y) = 6y + x^2 \quad h'(y) = 6y \quad h(y) = 3y^2$$

$$F(x,y) = x^2y + 3y^2 + x = C$$

$$\textcircled{8} \underbrace{(ye^x + e^y)}_{M(x,y)} dx + \underbrace{(e^x + xe^y)}_{N(x,y)} dy = 0$$

$$\frac{\partial M}{\partial y} = e^x + e^y = \frac{\partial N}{\partial x} = e^x + e^y \quad \checkmark \quad F(x,y) = ye^x + xe^y = C$$

$$F(x,y) = \int (ye^x + e^y) dx + h(y) = ye^x + xe^y + h(y)$$

$$\frac{\partial F}{\partial y} = e^x + xe^y + h'(y) = e^x + xe^y \quad h'(y) = 0 \quad h(y) = C$$

IV INTEGRATING FACTOR (EULER FACTOR)

→ If the differential equation

$$M(x,y)dx + N(x,y)dy = 0$$

with $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ can be made an exact dif. of the form

$$\mu(x,y) M(x,y)dx + \mu(x,y) N(x,y)dy = 0$$

$\mu(x,y) \rightarrow$ integrating factor

where μ is a suitable function of x and y so.

$$\left[\frac{\partial}{\partial y} (\mu M) = \frac{\partial}{\partial x} (\mu N) \right] \text{ "Exactness condition"}$$

$$\mu_y \cdot M + \mu \cdot M_y = \mu_x \cdot N + \mu N_x$$

HOW TO SOLVE?

Case 1 $\mu(x,y) = \mu(x)$ $\frac{M_y - N_x}{N} = f(x)$ $\frac{d\mu}{\mu} = f(x) dx$

or $\mu(x) = e^{\int f(x) dx}$

Case 2 $\mu(x,y) = \mu(y)$ $\frac{M_y - N_x}{-M} = g(y)$ $\frac{d\mu}{\mu} = g(y) dy$

or $\mu(y) = e^{\int g(y) dy}$

Case 3 If $\mu = \mu(x,y)$, the integrating factor must be given to us in some how.

(4) Solve obtained exact $\mu M dx + \mu N dy = 0$

Ex ① $(x^2 + y^2 + x) dx + xy dy = 0$

$M(x,y) = x^2 + y^2 + x$ $N(x,y) = xy$

$M_y = 2y \neq N_x = y$

Integrating factor Case ② $\frac{M_y - N_x}{-M} = \frac{2y - y}{-(x^2 + y^2 + x)} = \frac{y}{-(x^2 + y^2 + x)}$ not a function of one variable y

Case ① $\frac{M_y - N_x}{N} = \frac{2y - y}{xy} = \frac{1}{x}$

$\frac{d\mu}{\mu} = \frac{1}{x} dx$ $\ln(\mu(x)) = \ln x + \ln C$ $\mu(x) = Cx$ $C=1$ $\mu(x) = x$

Actually, $\int \frac{1}{x} dx = \ln|x|$ but because of C we ignore $|x|$ such that $\int \frac{1}{x} dx = \int \frac{y}{x^2 + y^2 + x} dx$ $\ln|x| = y^2 + C$
 $|x| = e^{y^2 + C}$ $x > 0$ $x = e^{y^2} \cdot e^C$ $x < 0$ $x = -e^{y^2} \cdot e^C$ } C this C contains all arbitrary variables.

$\mu(x) \cdot (x^2 + y^2 + x) dx + \mu(x) \cdot xy dy = 0$
 $\underbrace{(x^2 + y^2 + x)}_{M_1(x,y)} dx + \underbrace{xy}_{N_1(x,y)} dy = 0$

$\frac{\partial M_1}{\partial y} = 2xy = \frac{\partial N_1}{\partial x} = 2xy$ ✓

$F(x,y) = \int x^2 y dy + p(x)$

$F(x,y) = \frac{x^2 y^2}{2} + p(x)$ $\frac{\partial F}{\partial x} = xy^2 + p'(x) = x^2 + xy^2 + x^2$

$p'(x) = x^3 + x^2$ $p(x) = \frac{x^4}{4} + \frac{x^3}{3}$

$F(x,y) = \frac{x^2 y^2}{2} + \frac{x^4}{4} + \frac{x^3}{3} = C$

② Show that $\underbrace{(2yx e^x + 2x^2 y e^x)}_{M(x,y)} dx + \underbrace{(by^3 x - 2x^2 e^x)}_{N(x,y)} dy = 0$

is NOT EXACT but becomes exact when multiplied with

$\mu(x,y) = \frac{1}{xy^2}$ and solve diff.

$M_y = x e^x + 2x^2 e^x \neq N_x = by^3 - 4x e^x - 2x^2 e^x$ NOT EXACT

Multiply with $\frac{1}{xy^2}$

$$\frac{1}{xy^2} (2yx e^x + 2x^2 y e^x) dx + \frac{1}{xy^2} (by^3 x - 2x^2 e^x) dy = 0$$

$$\underbrace{\left(2 \frac{e^x}{y} + \frac{2x}{y} e^x \right)}_{P(x,y)} dx + \underbrace{\left(by - \frac{2x}{y^2} e^x \right)}_{Q(x,y)} dy = 0$$

$P_y = -\frac{2e^x}{y^2} - \frac{2x}{y^2} e^x = Q_x = -\frac{2}{y^2} e^x - \frac{2x}{y^2} e^x$ EXACT ✓ DIF

Obtain solutions by finding potential function $F(x,y)$

$$F(x,y) = \int \left(by - \frac{2x \cdot e^x}{y^2} \right) dy + h(x)$$

$$= by^2 + \frac{2}{y} x e^x + h(x)$$

$$\frac{\partial F}{\partial x} = + \frac{2}{y} (e^x + x e^x) + h'(x) = \frac{2}{y} e^x + \frac{2x}{y} e^x$$

$$h'(x) = 0 \quad h(x) = C$$

$$F(x,y) = by^2 + \frac{2}{y} x e^x = C$$

$$\textcircled{3} \quad \underbrace{\cos x \, dy}_{N(x,y)} - \underbrace{(2y \sin x - 3) \, dx}_{M(x,y)} = 0$$

$$N(x,y) = \cos x \quad M(x,y) = 3 - 2y \sin x$$

$$N_x = -\sin x \neq M_y = -2 \sin x \quad \text{NOT EXACT.}$$

$$\frac{M_y - N_x}{N(x,y)} = \frac{-\sin x}{\cos x} = -\tan x$$

$$\frac{d\mu}{\mu} = -\tan x \, dx$$

$$\ln \mu(x) = \ln |\cos x| + \ln C$$

$$\mu(x) = C |\cos x| \quad C=1$$

$$\mu(x) = \cos x$$

$$\underbrace{\cos^3 x \, dy}_{N(x,y)} - \underbrace{(2y \cos x \sin x - 3 \cos x) \, dx}_{M(x,y)} = 0$$

$$\frac{\partial N_1}{\partial x} = -2 \cos x \sin x$$

$$\frac{\partial M_1}{\partial y} = -2 \cos x \sin x$$

$$F(x,y) = \int \cos^3 x \, dy + p(x)$$

$$F(x,y) = \int \frac{\cos 2x + 1}{2} \, dy + p(x)$$

$$F(x,y) = \left(\frac{\cos 2x + 1}{2} \right) y + p(x)$$

$$\sqrt{\cos^3 x = \frac{\cos 2x + 1}{2}}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= \cos^2 x - (1 - \cos^2 x) \\ &= 2\cos^2 x - 1 \end{aligned}$$

$$\frac{\partial F}{\partial x} = \frac{y}{2} (-\sin 2x) + p'(x)$$

$$= -y \sin 2x + 3 \cos x$$

$$p'(x) = 3 \cos x \quad p(x) = 3 \sin x$$

$$\cos^2 x = \frac{\cos 2x + 1}{2}$$

$$F(x,y) = \frac{y}{2} (\cos 2x + 1) + 3 \sin x = C$$

EXERCISES

$$(1) \underbrace{(2xy^2 - 3y^3)}_{M(x,y)} dx + \underbrace{(7 - 3xy^2)}_{N(x,y)} dy = 0$$

$$M_y = 4xy - 9y^2$$

$$N_x = -3y^2$$

$$M_y - N_x = 4xy - 9y^2 + 3y^2 = 4xy - 6y^2$$

$$\frac{M_y - N_x}{-M} = - \frac{4xy - 6y^2}{2xy^2 - 3y^3} = -2 \frac{2xy - 3y^2}{y(2xy - 3y^2)} = -\frac{2}{y}$$

$$\frac{d\mu}{\mu} = -\frac{2}{y} dy \quad \ln \mu(y) = -2 \ln y \quad \mu(y) = \frac{C}{y^2} \quad C=1$$

$$\mu(y) = \frac{1}{y^2}$$

$$\underbrace{(2x - 3y)}_{M_1(x,y)} dx + \underbrace{\left(\frac{7}{y^2} - 3x\right)}_{N_1(x,y)} dy = 0$$

$$\frac{\partial M_1}{\partial y} = -3$$

$$\frac{\partial N_1}{\partial x} = -3$$

$$F(x,y) = \int (2x - 3y) dx + h(y) = x^2 - 3xy + h(y)$$

$$\frac{\partial F}{\partial y} = -3x + h'(y) = \frac{7}{y^2} - 3x$$

$$h'(y) = \frac{7}{y^2} \quad h(y) = -\frac{7}{y}$$

$$F(x,y) = x^2 - 3xy - \frac{7}{y} = C$$

$$(2) (y^2 + 2x^2) dx + xy dy = 0$$

$$\frac{dy}{dx} = -\frac{y^2 + 2x^2}{xy} = -\left[\frac{y}{x} + \frac{2x}{y}\right] \rightarrow \text{homogeneous not separable}$$

$$M(x,y) = y^2 + 2x^2$$

$$N(x,y) = xy$$

$$M_y = 2y$$

$$N_x = y$$

$$M_y - N_x = 2y - y = y$$

$$\frac{M_y - N_x}{N} = \frac{y}{xy} = \frac{1}{x}$$

$$\frac{d\mu}{\mu} = \frac{dx}{x}$$

$$\ln \mu(x) = \ln x + \ln C \quad \mu(x) = Cx \quad \mu(x) = x$$

$$\mu(x) = Cx \quad \mu(x) = x$$

$$\underbrace{x(y^2 + 2x^2)}_{M_1} dx + \underbrace{x^2 y}_{N_1} dy = 0$$

$$\frac{\partial M_1}{\partial y} = 2xy = \frac{\partial N_1}{\partial x} = 2xy$$

$$F(x,y) = \int x^2 y dy + g(x) = \frac{x^2 y^2}{2} + g(x)$$

$$\frac{\partial F}{\partial x} = xy^2 + g'(x) = xy^2 + 2x^3 \quad g'(x) = 2x^3 \quad g(x) = \frac{x^4}{2} + C$$

$$F(x,y) = \frac{x^2 y^2}{2} + \frac{x^4}{2} = C$$

③ $y dx + (y^2 - x) dx = 0$

NOT SEPARABLE $y dx$

NOT HOMOGENEOUS $y^2 - x$ 2,1

$$M(x,y) = y$$

$$M_y = 1$$

$$N(x,y) = y^2 - x$$

$$N_x = -1$$

$$M_y - N_x = 2$$

$$\frac{M_y - N_x}{-N} = -\frac{2}{y}$$

$$\frac{dM}{N} = -\frac{2}{y} dy$$

$$h(y) = -2 \ln y + C$$

$$M(y) = \frac{C}{y^2} \quad C=1 \quad M(y) = \frac{1}{y^2}$$

$$\underbrace{\frac{1}{y}}_{M_1} dx + \underbrace{\left(1 - \frac{x}{y^2}\right)}_{N_1} dx = 0$$

$$\frac{\partial M_1}{\partial y} = -\frac{1}{y^2} = \frac{\partial N_1}{\partial x} = -\frac{1}{y^2}$$

$$F(x,y) = \int \frac{1}{y} dx + h(y) = \frac{x}{y} + h(y)$$

$$\frac{\partial F}{\partial y} = -\frac{x}{y^2} + h'(y) = 1 - \frac{x}{y^2} \quad h'(y) = 1 \quad h(y) = y$$

$$F(x,y) = \frac{x}{y} + y = C$$

$$\textcircled{4} \quad \underbrace{y dx}_{M(x,y)} + \underbrace{(4x - y^2) dy}_{N(x,y)} = 0$$

$$M(x,y) = y \quad N(x,y) = 4x - y^2 \quad M_y = 1 \neq N_x = 4$$

$$\frac{M_y - N_x}{-M} = \frac{1-4}{-y} = \frac{-3}{-y} = \frac{3}{y}$$

$$\frac{d\mu}{\mu} = \frac{3}{y} dy \quad \ln|\mu(y)| = 3 \ln|y| + \ln C \quad \mu(y) = y^3$$

$$\underbrace{y^3 y dx}_{M_1} + \underbrace{y^3 (4x - y^2) dy}_{N_1} = 0$$

$$\frac{\partial M_1}{\partial y} = 4y^3 \quad \frac{\partial N_1}{\partial x} = 4y^3$$

$$F(x,y) = \int y^4 dx + g(y) = y^4 x + g(y)$$

$$\frac{\partial F}{\partial y} = 4y^3 x + g'(y) = 4xy^3 - y^5 \quad g'(y) = -y^5 \quad g(y) = -\frac{y^6}{6}$$

$$F(x,y) = y^4 x - \frac{y^6}{6} = C$$

$$\textcircled{5} \quad \underbrace{\sin(2y) dx}_{M(x,y)} + \underbrace{2x dy}_{N(x,y)} = 0$$

$$M_y = 2\cos(2y) \neq N_x = 2 \quad \text{NOT EXACT}$$

$$\frac{M_y - N_x}{-M} = \frac{2\cos(2y) - 2}{-\sin 2y} = \frac{2(1 - 2\sin^2 y) - 2}{-2\sin y \cos y} = 2 \tan y$$

$$\mu(y) = e^{\int 2 \tan y dy} = e^{-\ln|\cos y|} = \frac{1}{\cos^2 y}$$

$$\frac{1}{\cos^2 y} \cdot \sin(2y) dx + \frac{2x}{\cos^2 y} dy = 0 \quad \tan y dx + \frac{x}{\cos^2 y} dy = 0$$

$$\frac{\partial F}{\partial x} = \tan y \quad F(x,y) = x \tan y + h(y)$$

$$\frac{\partial F}{\partial y} = \frac{x}{\cos^2 y} + h'(y) = \frac{x}{\cos^2 y} \quad h'(y) = 0 \quad h(y) = C$$

$$F(x,y) = x \tan y = C$$

DEFINITION OF LINEAR EQUATION

Def. The O.D.E $F(x, y, y', \dots, y^{(n)}) = 0$ is linear if F is a linear function of variables $y, y', \dots, y^{(n)}$.

That is, F does not contain the multiplication of y and its derivatives with themselves. That is, linear n th order D.E will be in the form:

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = p(x)$$

Moreover, if $p(x) = 0$ then D.E is called homogeneous.

If $p(x) \neq 0$ then D.E is called nonhomogeneous.

Ex 1. $\frac{1}{x^2}y'' + \frac{2}{x}y' + e^x y = \underbrace{e^x \ln x}_{p(x)}$ linear 2nd order non-homogeneous D.E.

(2) $y'y'' + e^x y' = \ln x$ NON LINEAR

(3) $\sin y y''' + (2x^2 + 1)y'' + e^x y = \cos 2x$ NON LINEAR

(4) $2xy' + e^x y + (\cos y) = 0$ NON LINEAR

(5) $(y'')^2 + 2xy \neq \ln x$ NON LINEAR

VI LINEAR EQUATION (FIRST ORDER FIRST DEGREE LIN. EQUATION)

Definition An equation that can be reduced to the

$$\frac{dy}{dx} + p(x)y = q(x) \quad x \in I$$

is called a linear first order equation. We assume $p(x), q(x)$ are continuous on I .

NOTE (1) Linear equation is the most used first order diff. equation in engineering problems.

(2) See that $a_1(x) \frac{dy}{dx} + a_0(x)y = b(x)$ $a_1(x) \neq 0$ can be reduced to form

$$\frac{dy}{dx} + \underbrace{\frac{a_0(x)}{a_1(x)}}_{p(x)} y = \underbrace{\frac{b(x)}{a_1(x)}}_{q(x)} \quad a_1(x) \neq 0$$
$$\boxed{\frac{dy}{dx} + p(x)y = q(x)}$$

Theorem The linear equation

$$\frac{dy}{dx} + p(x)y = q(x)$$

has an integrating factor $\mu(x) = e^{\int p(x) dx}$ one parameter family solution of this equation

$$\boxed{y = e^{-\int p(x) dx} \left(\int e^{\int p(x) dx} q(x) dx + C \right)}$$

HOW TO SOLVE ^{1st} ORDER LINEAR DIFF. EQUATIONS & OBTAIN INTEGRATION FACTOR

(1) Write diff equation in the form

$$\frac{dy}{dx} + p(x)y = q(x) \quad (1)$$

(2) Obtain integrating factor. $\mu(x)$

$$(\mu(x) \cdot y)' = \mu'(x)y + \mu(x)y' \quad (2)$$

Multiply both sides of (1) with $\mu(x)$

$$\mu(x) \cdot y' + \mu(x)p(x)y = \mu(x)q(x) \quad (3)$$

If condition is satisfied $\mu(x)$ is an integrating factor,
Left hand-side of (3) and right hand-side of (2) must be equal

$$\cancel{\mu(x)y'} + \mu(x)p(x)y = \mu'(x)y + \cancel{\mu(x)y'}$$

$$\mu'(x)y = \mu(x)p(x)y$$

$$\frac{\mu'}{\mu} = p(x)$$

$$\mu(x) = e^{\int p(x) dx}$$

$$\mu'(x) = \underbrace{e}_{\mu(x)} p(x)$$

(3) Multiply both sides with $\mu(x)$ such that left side will be equal to $\frac{d}{dx} [\mu(x)y]$

$$\mu(x) \frac{dy}{dx} + \mu(x)p(x)y = q(x)\mu(x)$$

$$\frac{d}{dx} [\mu(x)y] = \mu(x)q(x) \quad (4)$$

(4) Take integral of (4)

$$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)q(x) dx + C \right] \quad (5)$$

Ex. 1 $\frac{dy}{dt} + 2y = 3 \quad y(0) = 5$

$p(t) = 2 \quad q(t) = 3$

$\mu(t) = e^{\int 2 dt} = e^{2t}$

$e^{2t} \frac{dy}{dt} + 2 \cdot e^{2t} y = 3e^{2t}$

$(e^{2t} \cdot y)' = 3 \cdot e^{2t}$

$\int (e^{2t} y)' dt = \int 3e^{2t} dt \Rightarrow e^{2t} y = \frac{3}{2} e^{2t} + C$

$y(t) = \frac{3}{2} e^{2t} + C \cdot e^{-2t} \rightarrow \text{General solution}$

$y(0) = 5$ $y(0) = \frac{3}{2} \cdot e^0 + C = 5 \quad C = 5 - \frac{3}{2} = 7/2$

$y(t) = \frac{3}{2} e^{2t} + \frac{7}{2} e^{-2t} \rightarrow \text{Particular solution}$

Ex. 2 $y' + \frac{1}{x-2} y = x^2$

$p(x) = \frac{1}{x-2}$

$\mu(x) = e^{\int \frac{1}{x-2} dx} = e^{\ln|x-2|} = (x-2)$

we use $(x-2)$, because $-(x-2)$ or $e^{\ln|x-2|+C}$ has the same results: they are just positive constants
 See $-(x-2)y' - y = -(x-2)x^2 \quad (-(x-2)y)' = -(x^3 - 2x^2)$

$-(x-2)y = \frac{2x^3}{3} - \frac{x^4}{4} + C$

$y = \frac{\frac{x^4}{4} - \frac{2x^3}{3} - C}{x-2}$

$(x-2)y' + y = (x-2)x^2$

$((x-2)y)' = x^3 - 2x^2 \Rightarrow$

$(x-2)y = \frac{x^4}{4} - \frac{2x^3}{3} + C$

$\int ((x-2)y)' = \int (x^3 - 2x^2) dx$

$y(x) = \frac{\frac{x^4}{4} - \frac{2x^3}{3} + C}{(x-2)}$

Ex 3 $(1+x^2)dy + (2xy + \tan x)dx = 0$

$$\frac{dy}{dx} = -\frac{(2xy + \tan x)}{1+x^2}$$

$$\frac{dy}{dx} + \underbrace{\frac{2x}{1+x^2}y}_{P(x)} = -\underbrace{\frac{\tan x}{1+x^2}}_{Q(x)} \quad (\ln \cos x) = -\frac{1}{x}$$

$$\mu(x) = e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = (1+x^2)$$

$$(1+x^2) \frac{dy}{dx} + 2xy = -\tan x \quad \Rightarrow \quad \int ((1+x^2)y)' dx = \int \tan x dx$$

$$(1+x^2)y = \ln|\cos x| + C$$

$$((1+x^2)y)' = -\tan x$$

$$y(x) = \frac{\ln|\cos x|}{1+x^2} + \frac{C}{1+x^2}$$

EXERCISES

① $y' - 3y = e^x$ $\mu(x) = e^{-\int 3dx} = e^{-3x}$

$$(e^{-3x}y' - 3e^{-3x}y) = e^{-3x} \cdot e^x$$

$$(e^{-3x}y)' = e^{-2x} \quad \Rightarrow \quad \int (e^{-3x}y)' dx = \int e^{-2x} dx$$

$$e^{-3x}y = \frac{e^{-2x}}{-2} + C$$

$$y(x) = -\frac{e^x}{2} + C \cdot e^{3x}$$

② $y^2 dx + (3xy - 1)dy = 0$

$$\frac{dx}{dy} = \frac{y^2}{1-3xy} \quad \Rightarrow \quad \frac{dx}{dy} = \frac{1-3xy}{y^2}$$

$$\frac{dx}{dy} + \frac{3x}{y} = \frac{1}{y^2}$$

$$P(y) = e^{\int 3/y dy} = e^{3 \ln|y|} = y^3$$

$$y^3 \frac{dx}{dy} + y^2 \frac{3x}{y} = y^3 \cdot \frac{1}{y^2}$$

$$\int (y^3 x)' dy = \int y dy \quad \Rightarrow \quad y^3 x = \frac{y^2}{2} + C \quad x = \frac{y^2}{2y^3} + \frac{C}{y^3}$$

③ $y' - y \tan x = 0$ $y(0) = 1$ $q(x) = 0$ BOTH LINEAR AND SEPARABLE

$$\frac{dy}{dx} = y \tan x \quad \int \frac{dy}{y} = \int \tan x dx \quad \ln|y| = -\ln|\cos x| + \ln C$$

General Sol.

$$\ln|y(0)| = -\ln|\underbrace{\cos 0}_1| + \ln C \Rightarrow C = 1$$

$$\ln|y(x)| = -\ln|\cos x| + \ln 1 \quad y(x) = \frac{1}{|\cos x|} \rightarrow \text{Particular solution}$$

④ $y' - y e^x = 0$

$$\frac{dy}{dx} = y e^x$$

$$\frac{dy}{y} = e^x dx$$

$$\ln|y| = e^x + C_1$$

Both linear and sep.

$$|y(x)| = e^{e^x + C_1}$$

$$= e^{e^x} \cdot \underbrace{e^{C_1}}_C$$

$$|y(x)| = e \cdot C$$

⑤ $xy' + 2y = x \sin x$ $y(\pi) = 1$

$$y' + \frac{2}{x}y = \sin x \quad (\text{linear})$$

$$\mu(x) = e^{\int 2/x dx} = e^{2 \ln|x|} = x^2$$

$$x^2 y' + 2xy = x^2 \sin x \Rightarrow (x^2 y)' = x^2 \sin x$$

$$\int (x^2 y)' dx = \int x^2 \sin x dx$$

$$\int x^2 \sin x dx = -x^2 \cos x + \int 2x \cos x dx$$

$x^2 = u \quad \sin x dx = dv$

$$2x dx = du \quad v = -\cos x$$

$$\int x \cos x dx = x \sin x - \int \sin x dx$$

$$x = u \quad \cos x dx = dv$$

$dx = dv \quad \sin x = v$

$$\int x \cos x dx = x \sin x + \cos x$$

$$x^2 y = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

$$y(x) = -\cos x + \frac{2 \sin x}{x} + \frac{2 \cos x}{x^2} + \frac{C}{x^2}$$

General solution

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x$$

Particular solution

$$y(\pi) = 1 \Rightarrow y(\pi) = -\cos \pi + \frac{2 \sin \pi}{\pi} + \frac{2 \cos \pi}{\pi^2} + \frac{C}{\pi^2}$$

$$y(\pi) = 1 - \frac{2}{\pi^2} + \frac{C}{\pi^2} = 1 \quad C = 2$$

$$y(x) = -\cos x + \frac{2 \sin x}{x} + \frac{2 \cos x}{x^2} + \frac{2}{x^2} \quad y(\pi) = 1 \quad \text{particular}$$

(12)

$$(6) (x^2+1)y' + 2xy = x^2(x^2+1)$$

$$y' + \frac{2x}{x^2+1} = x^2$$

$$\mu(x) = e^{\int \frac{2x}{x^2+1} dx} = e^{\ln(x^2+1)} = x^2+1$$

$$(x^2+1)y' + 2xy = x^2(x^2+1)$$

$$((x^2+1)y)' = x^4+x^2$$

$$\int ((x^2+1)y)' dx = \int (x^4+x^2) dx \Rightarrow (x^2+1)y = \frac{x^5}{5} + \frac{x^3}{3} + C$$

$$y(x) = \frac{x^5}{5} + \frac{x^3}{3} + C$$

$$(7) y' + y \tan x = \sec x \quad y(0)=1$$

$$p(x) = \tan x \quad q(x) = \sec x$$

$$\mu(x) = e^{\int \tan x dx} = e^{-\ln(\cos x)} = \frac{1}{\cos x} = \sec x$$

$$\sec x y' + y \cdot \frac{\sin x}{\cos x} \cdot \sec x = \sec^2 x$$

$$\sec x y' + y \cdot \sin x \sec^2 x = \sec^2 x$$

$$(\sec x y)' = \sec^2 x$$

$$\sec x \cdot y = \tan x + C \quad y(x) = \frac{\tan x}{\sec x} + \frac{C}{\sec x} \rightarrow \text{General solution}$$

$$y(0)=1 \quad y(0) = \frac{\tan 0}{\sec 0} + \frac{C}{\sec 0} = 1 \quad C=1$$

$$(\sec x) y = \tan x + 1$$

(VI) BERNOULLI DIFF EQUATION (SPECIAL CASE OF LIN)

Def An equation of the form

$$\frac{dy}{dx} + p(x)y = q(x)y^n \text{ is called Bernoulli diff. eqn}$$

• $n=0$ $\frac{dy}{dx} + p(x)y = q(x) \rightarrow \text{Linear diff}$

• $n=1$ $\frac{dy}{dx} + p(x)y = q(x)y$ $\frac{dy}{dx} = y(q(x) - p(x))$

$$\frac{dy}{dx} = (q(x) - p(x))y \rightarrow \text{separable}$$

Multiply both sides with y^{-n}

$$y^{-n} \frac{dy}{dx} + p(x)y^{1-n} = q(x)$$

$$v = y^{1-n}$$

$$\frac{1}{1-n} \frac{dv}{dx} + p(x)v = q(x)$$

$$\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

$$\frac{dv}{(1-n)dx} = y^{-n} \frac{dy}{dx}$$

$$\frac{dv}{dx} + \underbrace{(1-n)p(x)v}_{P(x)} = \underbrace{(1-n)q(x)}_{Q(x)} \rightarrow \text{Linear Diff}$$

(*) For $n > 0$ $y=0$ is always solution

HOW TO SOLVE

(1) Write equation in the form

$$\frac{dy}{dx} + p(x)y = q(x)y^n \quad (1)$$

(2) Define a new variable

$$v(x) = y^{1-n}$$

(3) Convert system into a new system by dividing y^n

$$\frac{1}{y^n} \frac{dy}{dx} + p(x)y^{1-n} = q(x)$$

$$\frac{1}{1-n} \frac{dv}{dx} + p(x)v = q(x) \quad (2)$$

(4) Solve lin. equation given in (2)

(5) Obtain $y(x) = (v(x))^{1/(1-n)}$

(20)

Ex 1 $\frac{dy}{dx} + y = xy^3$

$n=3$ $\star y=0$ is a solution since $n=3>0$.

$$v = y^{1-n} = y^{1-3} = y^{-2} \quad v = \frac{1}{y^2} \quad \frac{dv}{dx} = -\frac{2}{y^3} \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{-y^3}{2} \frac{dv}{dx}$$

$$-\frac{y^3}{2} \frac{dv}{dx} + y = xy^3$$

$$\frac{dv}{dx} - \frac{2}{y^2} = -2x \Rightarrow \frac{dv}{dx} - 2v = -2x \rightarrow \text{linear}$$

$\rightarrow$ Fractional way $v = y^{-2} \quad \frac{dv}{dx} \left(-\frac{1}{2} \right) + v = x$

$\frac{1}{1-n}$

$$\mu(x) = e^{-2 \int dx} = e^{-2x}$$

$$e^{-2x} \frac{dv}{dx} - 2e^{-2x} v = -2e^{-2x} \cdot x$$

$$(e^{-2x} v)' = -2e^{-2x} \cdot x$$

$$\int (e^{-2x} v)' dx = \int -2e^{-2x} x dx$$

$$e^{-2x} v = x \cdot e^{-2x} + \frac{e^{-2x}}{2} + C$$

$$v(x) = \left(x + \frac{1}{2}\right) + Ce^{2x} \rightarrow \text{General sol. of lin.}$$

$$\left(\frac{1}{y(x)}\right)^2 = \left(x + \frac{1}{2}\right) + C \cdot e^{2x} \rightarrow \text{General sol. of Bernoulli}$$

Ex 2 $\frac{dy}{dx} + \frac{y}{2x} = \frac{x}{y^3}$

$n = -3 \quad v = y^{1-(-3)} = y^4$

$\frac{dv}{dx} = 4y^3 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{4y^3} \frac{dv}{dx}$

$\frac{1}{4y^3} \frac{dv}{dx} + \frac{y}{2x} = \frac{x}{y^3} \Rightarrow \frac{dv}{dx} + \frac{4y^4}{2x} = 4x$

$\frac{dv}{dx} + \frac{2}{x}v = 4x \rightarrow \text{linear diff}$

$\mu(x) = e^{\int 2/x dx} = e^{2 \ln x} = x^2$

$x^2 \cdot \frac{dv}{dx} + \cancel{x^2} \cdot \frac{2}{\cancel{x}} v = 4x^3$

$(x^2 v)' = 4x^3 \Rightarrow \int (x^2 v)' dx = \int 4x^3 dx$
 $x^2 v = x^4 + C$

$v = x^2 + \frac{C}{x^2} \rightarrow \text{Sol}$

$y^4 = x^2 + \frac{C}{x^2} \rightarrow \text{Sol}$
Be

(Solve Exercise 4)

Ex 3 $xy' = 2x^2y + y^3 \quad n=3 \quad y=0$ is a solution

$y' = \frac{2}{x}y + \frac{y^3}{x^3} \quad v = y^{1-3} = y^{-2}$

$y' - \frac{2}{x}y = \frac{y^3}{x^3} \Rightarrow \frac{dv}{dx} = -\frac{2}{y^3} \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{y^3}{2} \frac{dv}{dx}$

$-\frac{y^3}{2} \frac{dv}{dx} - \frac{2}{x}y = \frac{y^3}{x^3} \Rightarrow \frac{dv}{dx} + \frac{4}{x} \left(\frac{1}{y^2} \right) = -\frac{2}{x^3}$

$\frac{dv}{dx} + \frac{4}{x}v = -\frac{2}{x^3} \Rightarrow \mu(x) = e^{\int \frac{4}{x} dx} = e^{4 \ln x} = x^4$

$x^4 \frac{dv}{dx} + 4x^3 v = -2x \Rightarrow \int (x^4 v)' dx = \int -2x dx$
 $x^4 v = -x^2 + C \quad v(x) = -\frac{1}{x^2} + \frac{C}{x^4}$
General sol
Lin.
 $\frac{1}{y^2(x)} = -\frac{1}{x^2} + \frac{C}{x^4} \rightarrow \text{General sol}$
Ben.

EXERCISES

① $6y^2 dx - x(2x^3 + y) dy = 0$

$$6y^2 dx = x(2x^3 + y) dy$$

$$\frac{dx}{dy} = \frac{2x^4}{6y^2} + \frac{xy}{6y^2}$$

$$\frac{dx}{dy} - x \cdot \frac{1}{6y} = \frac{1}{3y^2} x^4 \Rightarrow \text{Bernoulli}$$

$n=4 > 0$ $x=0$ is a solution

$$v = x^{1-4} = x^{-3} \quad \frac{dv}{dy} = -3x^{-4} \frac{dx}{dy} \Rightarrow \frac{dx}{dy} = -\frac{x^4}{3} \frac{dv}{dy}$$

$$-\frac{x^4}{3} \frac{dv}{dy} - x \cdot \frac{1}{6y} = \frac{1}{3y^2} x^4$$

$$\frac{dv}{dy} + \frac{v}{2y} = -\frac{1}{y^2} \Rightarrow p(y) = e$$

Lin.

$$\int \frac{1}{2y} dy = \frac{1}{2} \ln y = e^{1/2}$$

$$y^{1/2} \frac{dv}{dy} + \frac{v}{2\sqrt{y}} = -\frac{1}{y^{3/2}} \Rightarrow (y^{1/2} v)' = -y^{-3/2}$$

$$\int (y^{1/2} v)' dy = - \int y^{-3/2} dy$$

$$y^{1/2} v = -\frac{y^{-1/2}}{-1/2} + C$$

$$v(y) = \frac{2}{y} + C \cdot y^{-1/2} \Rightarrow \text{Sol of Lin}$$

$$x^{-3} = \frac{2}{y} + \frac{C}{\sqrt{y}} \Rightarrow \text{Sol of Bern}$$

② $\frac{dy}{dx} - \frac{1}{3x} y = -2x^4 y^4$ $n=4$ $y=0$ is a solution

$$v = y^{1-4} = y^{-3}$$

$$\frac{dv}{dx} = -3y^{-4} \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{y^4}{3} \frac{dv}{dx}$$

$$-\frac{y^4}{3} \frac{dv}{dx} - \frac{1}{3x} y = -2x^4 y^4 \Rightarrow \frac{dv}{dx} + \frac{1}{x} \left(\frac{1}{y^3} \right) = 6x^4 \Rightarrow \text{L.H.S.}$$

$$\mu(x) = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$x v' + \frac{1}{x} \cdot v \cdot x = 6x^5 \quad (xv)' = 6x^5 \Rightarrow \int (xv)' dx = \int 6x^5 dx$$

$$x \cdot v = x^6 + C \Rightarrow v(x) = x^5 + \frac{C}{x} \rightarrow \text{Sol of L.H.S.}$$

$$\frac{1}{y^3} = x^5 + \frac{C}{x} \rightarrow \text{Sol of I.F.}$$

③ $y' + 3y = y^3 e^{-x}$ $n=3 > 0$ $y=0$ is a solution

$$v = y^{1-3} = y^{-2} \quad \frac{dv}{dx} = -\frac{2}{y^3} \frac{dy}{dx} \quad \frac{dy}{dx} = -\frac{y^3}{2} \frac{dv}{dx}$$

$$-\frac{y^3}{2} \frac{dv}{dx} + 3y = y^3 e^{-x}$$

$$\frac{dv}{dx} - 6 \cdot \left(\frac{1}{y^2} \right) = -2e^{-x} \rightarrow \text{L.H.S.} \quad \mu(x) = e^{\int -6 dx} = e^{-6x}$$

$$(e^{-6x} v)' = -2e^{-7x}$$

$$\int (e^{-6x} v)' dx = \int -2e^{-7x} dx$$

$$e^{-6x} v = \frac{2}{7} e^{-7x} + C \Rightarrow v(x) = \frac{2}{7} e^{-x} + C e^{6x} \rightarrow \text{Sol of L.H.S.}$$

$$\frac{1}{y^2} = \frac{2}{7} e^{-x} + C e^{6x} \rightarrow \text{Sol of I.F.}$$

(2)

④ $y' = \frac{1}{x}y + \frac{y^2}{x^2}$ $y(1)=1$

$\frac{dy}{dx} - \frac{1}{x}y = \frac{y^2}{x^2}$ $n=2>0$ $y=0$ is a solution

$v = y^{1-2} = y^{-1}$ $\frac{dv}{dx} = -\frac{1}{y^2} \frac{dy}{dx}$ $\frac{dy}{dx} = -y^2 \frac{dv}{dx}$

$-y^2 \frac{dv}{dx} - \frac{1}{x}y = \frac{y^2}{x^2} \Rightarrow \frac{dv}{dx} + \frac{1}{x} \left(\frac{1}{y}\right) = -\frac{1}{x^2}$

$\frac{dv}{dx} + \frac{1}{x}v = -\frac{1}{x^2}$ $\mu(x) = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$

$x \cdot \frac{dv}{dx} + x \cdot \frac{1}{x} v = -\frac{x}{x^2} \Rightarrow \left(x \cdot \frac{dv}{dx} + v\right) = -\frac{1}{x^2}$

$(x \cdot v)' = -\frac{1}{x^2} \Rightarrow xv = \frac{1}{x} + C$

$v(x) = \frac{1}{x^2} + \frac{C}{x} \rightarrow$ Gen. of Lin.
 $\frac{1}{y(x)} = \frac{1}{x^2} + \frac{C}{x} \rightarrow$ Gen. of Ber.

$y(1)=1 \Rightarrow \frac{1}{1} = 1+C \Rightarrow C=0 \Rightarrow \frac{1}{y} = \frac{1}{x^2}$

⑤ $\frac{dx}{dy} + 2 \tan y x = (2 \cos y) x^{1/2}$ $n=1/2>0$ $x=0$ is a solution

$v = x^{1-1/2} = x^{1/2}$ $\frac{dv}{dy} = \frac{1}{2\sqrt{x}} \cdot \frac{dx}{dy}$ $\frac{dx}{dy} = 2\sqrt{x} \cdot v'$

$2\sqrt{x} v' + 2 \tan y x = 2 \cos y \sqrt{x}$

$v' + \tan y \sqrt{x} = \cos y \rightarrow$ lin.

$\mu(y) = e^{\int \tan y dy} = e^{-\ln \cos y} = 1/\cos y$

$\frac{1}{\cos y} v' + \frac{\sin y}{(\cos y)^2} v = 2 \cos y$

$\left(\frac{1}{\cos y} v\right)' = 2 \cos y \Rightarrow \frac{v}{\cos y} = 2 \sin y + C$

$v = 2 \sin y \cos y + C \cos y \rightarrow$ Gen. of Lin.
 $x^{1/2} = 2 \sin y \cos y + C \cos y$

EXISTENCE AND UNIQUENESS THEOREM

Question (Page 7) How can we say something about the solution without solving the D.E.

Answer ① By looking at the graph of solution (direction fields) ✓
② Existence and Uniqueness Theorem ?

EXISTENCE AND UNIQUENESS THEOREMS

Existence and Uniqueness Thm for I.V.P.

① Consider the I.V.P $\begin{cases} y' = f(x, y) \\ y(x_0) = y_0 \end{cases}$

a) (Existence) If $f(x, y)$ is continuous on a rectangle R containing (x_0, y_0)
 $R = \{ (x, y) \mid |x - x_0| < a, |y - y_0| < b \}$

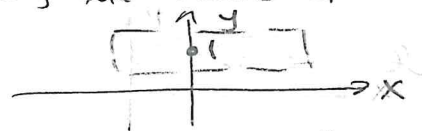
then, the I.V.P has "at least one solution" on a small interval $|x - x_0| < h$ ($h > 0$ small number)

b) Uniqueness If $\frac{\partial f}{\partial y}$ is continuous on R , then the I.V.P has a unique solution passing through (x_0, y_0)

Ex $\begin{cases} y' = y \\ y(0) = 1 \end{cases}$ ① $f(x, y) = y$ is continuous for $\forall y \in \mathbb{R}$ and $\forall x \in \mathbb{R} \therefore$
 $\frac{\partial f}{\partial y} = 1$ is also continuous for $\forall x, y \in \mathbb{R}$ ②

For ① So, it is possible to draw a rectangle R containing $(0, 1)$ such that f is continuous. Thus, $\exists$ at least one solution.

For ② Since $\frac{\partial f}{\partial y} = 1$ is also continuous $\forall x, y \in \mathbb{R}$, we have a unique solution for I.V.P.



max possible $R = (-\infty, \infty) \times (-\infty, \infty) = \mathbb{R}^2$.

Ex 7 $y' = x y^{1/2}$ $y \geq 0$ $x \in \mathbb{R}$

(a) $y(0) = 0$

(b) $y(0) = 1$

(a) $y(0) = 0$ Existence $f(x, y) = x y^{1/2}$ is continuous $\forall y \geq 0$ and $\forall x \in \mathbb{R}$
 so there exists at least one solution
 by E & U Theorem.

Uniqueness $\frac{\partial f}{\partial y} = x \frac{1}{2} y^{-1/2} = \frac{1}{2} \frac{x}{\sqrt{y}}$ is cont $\forall y > 0$ and $\forall x \in \mathbb{R}$.

Since $(0, 0)$ is on $y = 0$, there is no $\mathbb{R}$ containing $(0, 0)$ in which $\frac{\partial f}{\partial y}$ is continuous. So E & U Thm fails. That is, it says nothing about the uniqueness of solution.

So find another way to determine whether there is a unique solution or not.

So solve D.E $y' = x y^{1/2}$ "separable"

$$\int \frac{dy}{y^{1/2}} = \int x dx \quad \xrightarrow{\text{if } y \neq 0}$$

$$\frac{y^{1/2}}{1/2} = \frac{x^2}{2} + C_1 \Rightarrow y = \left(\frac{x^2}{4} + \left(\frac{1}{2} C_1 \right)^2 \right)^2$$

$$\Rightarrow y = \left(\frac{x^2}{4} + C \right)^2$$

Initial condition $y(0) = 0 \Rightarrow 0 = \left(\frac{0^2}{4} + C \right)^2 \Rightarrow C = 0$

Thus, $y = \frac{x^4}{16}$ is a solution for I.V.P.

On the other hand check $y(x) = 0$ $y(x) = 0$ satisfies $y' = x y^{1/2}$
 since $0' = 0$

And moreover, since $y(x) = 0$ $y(0) = 0$ satisfying I.C.

(••) I.C has at least two solutions $y_1(x) = \frac{x^4}{16}$ and $y_2(x) = 0$
 NOT UNIQUE

(b) $y(0) = 1$

Existence $f(x) = xy^{1/2}$ cont. $\forall y \geq 0 \quad \forall x \in \mathbb{R}$

So $\exists$ at least one solution. Since we can find a $\mathcal{R}$ containing $(0,1)$ such that f is continuous.

Uniqueness $\frac{\partial f}{\partial y} = \frac{x}{2\sqrt{y}}$ is continuous $\forall y > 0, \forall x \in \mathbb{R}$, then the given

$\mathcal{R}$ is suitable to have $\frac{\partial f}{\partial y}$ is continuous inside.

By E & U Theorem I.V.P with I.C $y(0) = 1$ has a unique solution passing $(0,1)$.

Ex What is max rectangular region to apply E & U Thm

$$y' = \frac{3x^2 + 4x + 2}{2(y-1)}$$

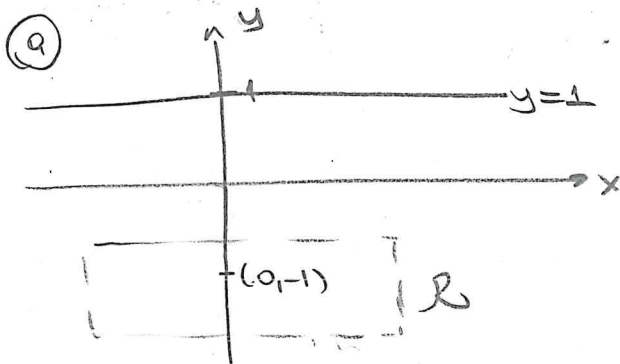
$f(x,y)$

(b) $y(0) = 1$

(a) $y(0) = -1$

E & U Thm $f(x,y)$ is cont. $\forall y \neq 1 \quad \forall x \in \mathbb{R}$.

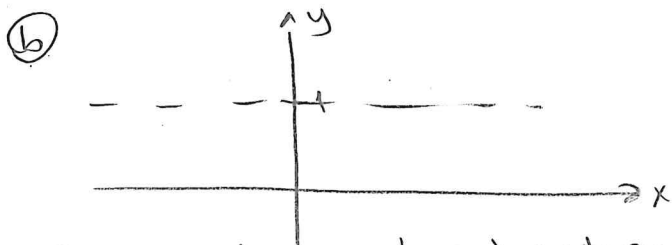
$$\frac{\partial f}{\partial y} = - \frac{3x^2 + 4x + 2}{2} \frac{1}{(y-1)^2} \text{ is cont. } \forall y \neq 1 \quad \forall x \in \mathbb{R}$$



So since it is possible to draw a rectangle $\mathcal{R}$ containing $(0,-1)$ by E & U Thm solution exists and is unique.

max $\mathcal{R} = (-\infty, 0] \times (-\infty, 1)$

$y=1$ is under for $y=1$



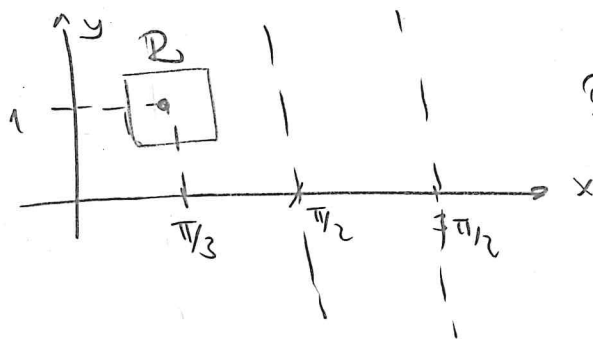
Since conditions of E & U Thm is not satisfied, Thm fails.

On the other hand when we solve this sep. equation we obtain two solutions

$$y = 1 \pm \sqrt{x^3 + 2x^2 + 2x} \quad (28)$$

Ex $y' = \frac{1}{\tan(t) \cdot \sqrt{y}}$ Find the largest possible $\mathcal{D}$ about $\lim_{t \rightarrow \pi/3} f(t, y)$ where $(\pi/3, 1) \in \mathcal{D}$ Thm is applicable.

$f(t, y)$ is cont $\forall y > 0$ $t > 0$ and $\cos t \neq 0 \Rightarrow t \neq (2k+1)\frac{\pi}{2}$
 $t \neq \frac{\pi}{2}$ $k = 0, \pm 1, \pm 2, \dots$



$\frac{\partial f}{\partial y} = \frac{1}{\tan t} \cdot \frac{1}{2\sqrt{y}}$ is cont. when $y > 0$ and with the same t values given above.

Then, max $\mathcal{D}$ containing $(\pi/3, 1)$ is $= (1, \pi/2) \times (0, \infty)$

Existence and Uniqueness Thm For Linear ODE

If the functions p and q are continuous on an interval I containing x_0 , then $\exists$ a unique soln. for the

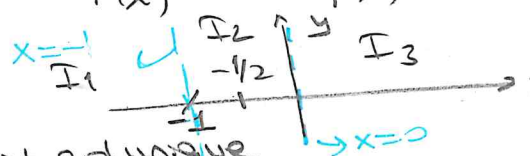
I.V.P $y' + p(x)y = q(x)$ in the interval I .

(General case $y' = \frac{q(x) - p(x)y}{r(x, y)}$ previous E&U Thm satisfied for all y)
 Ex Without solving obtain the largest solution interval of following D.I.E

$$\textcircled{1} \quad x \frac{dy}{dx} + \frac{2x+1}{x+1} y = x-1 \Rightarrow \frac{dy}{dx} + \underbrace{\frac{2x+1}{x(x+1)}}_{p(x)} y = \underbrace{\frac{x-1}{x}}_{q(x)}$$

$p(x)$ is cont when $x \neq 0, -1$

$q(x)$ is cont when $x \neq 0$



Let $y(-1) = -1$, then solution interval $I_3 = (0, \infty)$

Let $y(-1/2) = 0$, solution exist and unique in $I_2 = (-1, 0)$

Let $y(-2) = 3$, sol. exist and unique in $I_1 = (-\infty, -1)$

Let's solve $\frac{dy}{dx} + \frac{2x+1}{x(x+1)}y = \frac{x-1}{x}q(x)$

$$\mu(x) = e^{\int p(x) dx} = e^{\int \frac{2x+1}{x^2+x} dx} = e^{\ln(x^2+x)} = x^2+x$$

Multiply D.E with $\mu(x) = x^2+x$
 $(x^2+x) \cdot \frac{(x-1)}{x} = x^2-1$

$$(x^2+x)y' + (2x+1)y = x^2-1$$

$$(x^2+x)y = \int (x^2-1) dx$$

$$y(x) = \left(\frac{x^3}{3} - x + C \right) \cdot \frac{1}{x^2+x}$$

$$y(x) = \frac{x^3-3x}{3(x^2+x)} + \frac{C}{x^2+x}$$

$\underbrace{\hspace{1cm}}_{y_P} \qquad \underbrace{\hspace{1cm}}_{y_h}$

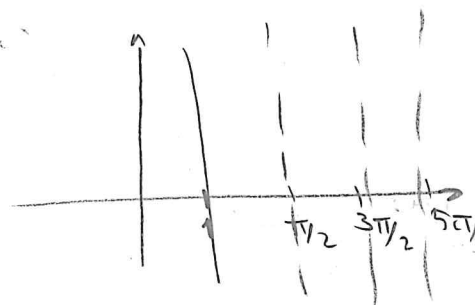
Let $\underline{y(1) = -1}$ $-1 = -\frac{2}{6} + \frac{C}{2} \Rightarrow C = -4/3$

I.V.P has solution $y(x) = \frac{x^3-3x}{3(x^2+x)} - \frac{4}{3(x^2+x)}$

Ex $\cos x y' = \sin x y - \sqrt{x-1} \quad y(3/2) = 0$

$$y' - \frac{\sin x}{\cos x} y = \frac{-\sqrt{x-1}}{\cos x} q(x)$$

$\underbrace{\hspace{1cm}}_{P(x)} \qquad \underbrace{\hspace{1cm}}_{q(x)}$



P is cont. when $\cos x \neq 0 \Rightarrow x \neq (2k+1)\frac{\pi}{2} \quad k=0, \pm 1, \dots$

q is cont. when $\cos x \neq 0$ and $x \geq 1$

It has a unique solution on $I = [1, \pi/2)$

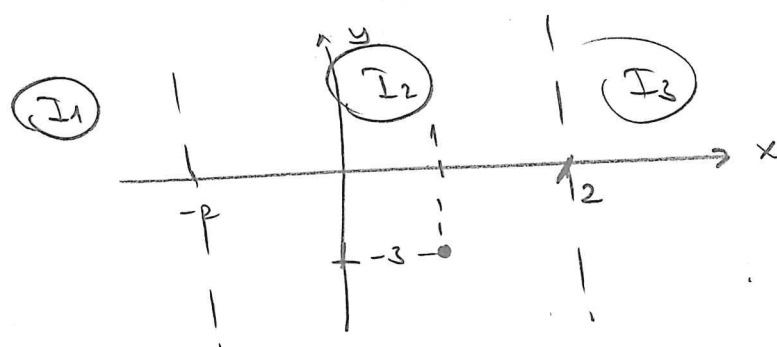
$$L = [1, \pi/2) \times (-\infty, \infty)$$

DIFFERENCES BETWEEN LINEAR & NONLINEAR EQUATIONS

- ① The interval of existence of a solution of a nonlinear equation can't be predicted, while for linear equations it is known in advance by E & U Theorem

Ex $y' + \underbrace{\frac{2x}{4-x^2} y}_{p(x)} = \underbrace{\frac{3x^2}{4-x^2}}_{q(x)} \quad y(1) = -3 \text{ Linear}$

E & U p, q are continuous when $4-x^2 \neq 0 \Rightarrow x \neq \pm 2$



$I_1 = (-\infty, -2)$
 $I_2 = (-2, 2) \rightarrow$ interval of solution for I.V.P. which contains I.C. $y(1) = -3$.
 $I_3 = (2, \infty)$

NOTE Without solving the D.E we predict the interval of solution.

Ex $y' = \underbrace{2xy^2}_f \quad y(0) = 4 \text{ nonlinear}$

$f(x, y) = 2xy^2$ Cont $\forall x, y \in \mathbb{R}$

$\frac{\partial f}{\partial y} = 4xy$ Cont $\forall x, y \in \mathbb{R}$

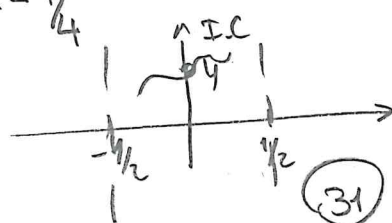
Thus, $\exists$ a unique solution in a rectangle containing $(0, 4)$

Thus, $\exists$ a unique solution in some interval $-h < x - x_0 < h \in (-\infty, \infty)$

What is h ? we can't predict h without solving D.E.

Let's solve D.E $\frac{y'}{y^2} = 2x \Rightarrow y = -\frac{1}{x^2 + C}$

$y(x) = -\frac{1}{x^2 + \frac{1}{4}}$ is defined if $x^2 - \frac{1}{4} \neq 0 \Rightarrow x \neq \pm \frac{1}{2}$
 $x \in (-\frac{1}{2}, \frac{1}{2})$



② General solution is usually called a solution containing all possible solutions for the D.E and contains a constant. Any solution which can not obtain from general solution is called singular solution

Ex $y' = y^2$ (Nonlinear eq).

$y' = y^2$ $y(x) = 0$ is a solution which cannot be obtained from the solution $y = -\frac{1}{x+C}$
 $\xrightarrow{\text{Sol 1}}$

$$\frac{dy}{y^2} = dx \Rightarrow -\frac{1}{y} = x+C \quad y = -\frac{1}{x+C} \quad \xrightarrow{\text{Sol 2}}$$

Remark For nonlinear equations one can obtain a solution containing a constant C , but this solution may not give all solutions. However, this is not the case for linear D.E.

③ Principle of Superposition

Linear equations obey the principle of Superposition. That is for homogeneous linear equations, linear combinations of solutions is also a solution of D.E

Suppose y_1, y_2 are solutions of a linear hom. D.E.

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = 0, \text{ then}$$

$\alpha y_1(x) + \beta y_2(x) \rightarrow$ linear combination of $y_1(x)$ and $y_2(x)$, $\alpha, \beta \in \mathbb{R}$ is also a solution. since

$$\begin{aligned} & a_n(x)(\alpha y_1^{(n)} + \beta y_2^{(n)}) + a_{n-1}(x)(\alpha y_1^{(n-1)} + \beta y_2^{(n-1)}) + \dots + a_0(x)(\alpha y_1 + \beta y_2) \\ &= \alpha (a_n(x)y_1^{(n)} + a_{n-1}(x)y_1^{(n-1)} + \dots + a_0(x)y_1) + \beta (a_n(x)y_2^{(n)} + \dots + a_0(x)y_2) \\ &= 0 \quad \text{since } y_1 \text{ and } y_2 \text{ are solutions} \end{aligned}$$

Therefore, $\alpha y_1 + \beta y_2$ is also a solution.

If it is $P(x)$ instead of 0 then for $\alpha y_1 + \beta y_2$ we must have $\alpha + \beta = 1$ Check please

Consider Nonlinear Case

$$y' = y^2 \Rightarrow y = -\frac{1}{x+C}$$

Let $y_1 = -\frac{1}{x}$ $y_2 = -\frac{1}{x+1}$ are solutions of $y' = y^2$

But $\alpha y_1 + \beta y_2 = -\frac{\alpha}{x} - \frac{\beta}{x+1}$ is not a solution of

$y' = y^2$ unless $\alpha = 0, \beta = 1$ and $\alpha = 1, \beta = 0$

$$(\alpha y_1 + \beta y_2)' = -\frac{\alpha}{x^2} - \frac{\beta}{(x+1)^2} \neq \left(-\frac{\alpha}{x} - \frac{\beta}{x+1}\right)^2 \quad \text{so not satisfying D.E.}$$

